MZUZU UNIVERSITY

FACULTY OF SCIENCE, TECHNOLOGY AND INNOVATION

DEPARTMENT OF INFORMATION AND COMMUNICATION TECHNOLOGY

BACHELOR OF SCIENCE IN INFORMATION AND COMMUNICATION TECHNOLOGY

BICT 4701: SOFTWARE ENGINEERING

END OF SEMESTER EXAMINATION

DATE: SEPTEMBER 2021 **TIME ALLOWED: 3 HOURS**

INSTRUCTIONS

i. There are **7** questions in this paper. **ANSWER ALL.**

i. There are **THREE** pages. **PLEASE CHECK**

iii. Marks are indicated for each question.

1. Discuss the following terms:
   a. Software process. **[2 Marks]**
   b. Use case diagrams. **[2 Marks]**
   c. Sequence diagrams. **[2 Marks]**
   d. Batch processing systems. **[2 Marks]**
   e. Embedded control systems. **[2 Marks]**

2. Draw a sequence diagram modeling the interactions involved when a student register for a course at Mzuzu University. Assume that courses may have limited enrolment, so the registration process must include checks that places are available. The student also needs to be authorized to access an online course catalog to find out about available courses. **[8 Marks]**

3. Systems Development Life Cycle(SDLC) involves a set of activities and technologies that go into producing an online system solution to an organizational problem or opportunity.

   a. Discuss the **five** phases of the SDLC and mention **one** major deliverable from each of the phases as discussed in class? **[10 Marks]**
   b. What are the **five** required skills of the systems analyst? **[5 Marks]**
   c. List the **seven** major activities that must be done during the design phase. **[7 Marks]**
   d. Explain **five** activities that are required to get the project organized during the project planning phase? **[5 Marks]**

4. Mzuzu Central Hospital decides to conduct online insulin pump control system for all diabetic patients to maintain blood glucose control. You are task to develop software for the same which has to meet the following requirements:

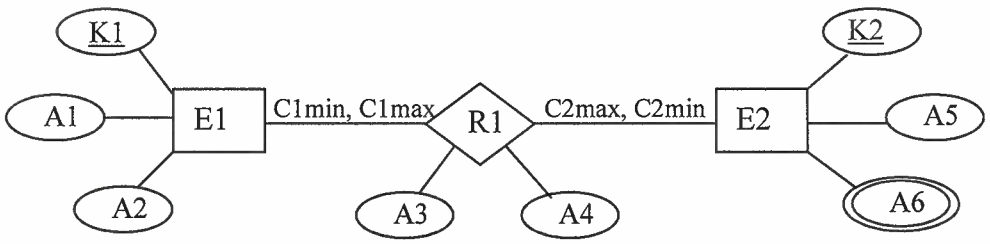
   i) Collects data from a blood sugar sensor and calculates the amount of insulin required to be injected.
   ii) Perform calculation based on the rate of change of blood sugar levels.
   iii) Sends signals to a micro-pump to deliver the correct dose of insulin.
   iv) Safety-critical system as low blood sugars can lead to brain malfunctioning, coma and death; high-blood sugar levels have long-term consequences such as eye and kidney damage.

   a. Prepare a work plan that will be analyzed to determine the overall project timeframe, costs, personnel requirements and software requirements? **[10 Marks]**
   b. Explain how you will conduct your feasibility study for the project. **[8 Marks]**
   c. Explain the role of Joint Application Development (JAD) in this the project **[4 Marks]**
   d. Draw an activity diagram that models the data processing in the insulin pump? **[10 Marks]**

5. Consider your experience with a Bank ATM Machine, draw a use case diagram that shows the data processing involved when you withdraw from the ATM machine? **[10 Marks]**

6. Mphatso has a large DVD movie collection. Her friends like to borrow her DVD's, and she needs a way to keep track of who has what. She maintains a list of friends, identified by unique FID's (friend identifiers) and a list of DVD's, identified by DVDID's (DVD identifiers). With each friend is the name and the all-important telephone numbers which she can call to get the DVD back. With each DVD is the star actor name and title. Whenever a friend borrows a DVD, Mphatso will enter that fact into her database along with the date borrowed. Whenever the DVD gets returned that fact too gets noted along with the date returned. Mphatso wants to keep a complete history of her friends' borrowing habits so that she can ask favors of the heavy borrowers (or perhaps refuse to make further loans to those who habitually don't return them quickly).

a. Below is an E-R diagram for a database to help Mphatso out. Provide appropriate names for entities E1, E2; attributes A1, A2,....A5; multi-value attribute A6, relationship R1, and cardinality constraint C1min, C1max, C2min and C2max.



Entity E1:_____ Key K1:_____ Attribute A1: _____ Attribute A2:_____

Entity E2:_____ Key K2:_____ Attribute A5: _____ Multi-value Attribute A6:_

Relationship R1:_____ Attribute A3:_____ Attribute A4:_____

Cardinality C1min: _____ C1max:_____ C2max :_____ C2min :_____

[15 Marks]

7. Discuss each of the following types of software maintenance:

a. Corrective. **[2 Marks]**

b. Adaptive. **[2Marks]**

c. Perfective. **[2 Marks]**

d. Preventive. **[2 Marks]**

END OF QUESTION PAPER